Abstract

This project aims to design and implement an innovative PaaS platform for one-click deployment of web applications using a public or private GitHub link via secure OAuth authentication. Deployment takes place in a cloud environment orchestrated by Kubernetes, which manages Docker container orchestration to ensure scalability and high availability (HA) with horizontal autoscaling based on load metrics. The platform adopts a siloed multi-tenancy strategy to fully isolate users' resources, thereby enhancing security and data privacy. The user experience centers on an intuitive, responsive web interface that allows repository submission for automatic builds as well as real-time monitoring of runtime logs and performance metrics (CPU, memory, network traffic). It also supports dynamic subdomain generation with automatic SSL certificate issuance via Let's Encrypt for secure accessibility. Furthermore, the platform integrates advanced environment management including configuration of environment variables and secure handling of sensitive secrets (such as API keys or passwords) stored encrypted in Kubernetes Secrets to prevent accidental exposure. A deployment history is maintained, offering rollback capabilities to previous versions in case of errors, along with detailed logs of builds, tests, and production deployments for complete traceability. Data persistence is provided via Kubernetes PersistentVolumes, enabling stateful applications (for example, those using databases) to retain data across redeployments and scaling events. Inspired by solutions like Railway, Render, and Vercel, this system supports multiple languages and frameworks (e.g., Node.js, Python), integrates automated CI/CD pipelines for continuous updates via GitHub webhooks, and facilitates agile DevOps adoption by reducing time-to-production and optimizing cloud costs through dynamic resource allocation, making the platform suitable for hybrid or multi-cloud environments.

Keywords: PaaS, Automated deployment, GitHub, Git, Cloud computing, Kubernetes, High availability, SSL.